

Phase 4 Causal Transfer Runbook

This runbook operationalizes the phase-4 goal: strengthen the causal interpretation of the transfer result before adding new curriculum mechanisms.

For the conceptual protocol, use docs/PHASE_4_CAUSAL_TRANSFER_PROTOCOL.md.

Step 1: Choose The Paired Recipes

Required comparison:

1. rotating_opponents_holdout_endpoint
2. rotating_plus_nemesis_novelty_replay

Optional third arm:

3. rotating_plus_replay_aware_selection

Step 2: Generate Phase-4-Labeled Transfer Configs

These commands keep the transfer setup the same while stamping the configs with study_phase=phase_4. The archived config and run-root names stay aligned with the existing runs/transfer_suite convention rather than introducing a separate phase_4_ path prefix.

```
python build_transfer_suite.py --recipe rotating_opponents_holdout_endpoint --study-phase phase_4
--output configs/transfer_suite/rotating_opponents_holdout_endpoint.json --output-root
runs/transfer_suite/rotating_opponents_holdout_endpoint
python build_transfer_suite.py --recipe rotating_plus_nemesis_novelty_replay --study-phase phase_4
--output configs/transfer_suite/rotating_plus_nemesis_novelty_replay.json --output-root
runs/transfer_suite/rotating_plus_nemesis_novelty_replay
python build_transfer_suite.py --recipe rotating_plus_replay_aware_selection --study-phase phase_4
--output configs/transfer_suite/rotating_plus_replay_aware_selection.json --output-root
runs/transfer_suite/rotating_plus_replay_aware_selection
```

Skip the third command if you are not running the optional replay-aware arm.

Step 3: Run Paired Replicates

Use identical offsets for every recipe.

If you are extending the current five-offset dataset, keep:

- a: 0
- b: 1000
- c: 2000
- d: 3000
- e: 4000

If you want the stronger phase-4 target, add:

- f: 5000
- g: 6000
- h: 7000

- i: 8000
- j: 9000

Example commands for the required two-arm comparison:

```
python run_suite.py --config configs\transfer_suite\rotating_opponents_holdout_endpoint.json
--seed-offset 0 --replicate-label a
python run_suite.py --config configs\transfer_suite\rotating_plus_nemesis_novelty_replay.json
--seed-offset 0 --replicate-label a
python run_suite.py --config configs\transfer_suite\rotating_opponents_holdout_endpoint.json
--seed-offset 1000 --replicate-label b
python run_suite.py --config configs\transfer_suite\rotating_plus_nemesis_novelty_replay.json
--seed-offset 1000 --replicate-label b
```

Continue the same pattern for the remaining offsets. If you include the optional third arm, run it on the same offsets too.

If a replicate label is rerun because of an execution-quality outlier, keep the cleaner rerun in the active recipe root and move the superseded original under runs/transfer_suite/_superseded/ <recipe> / while preserving its original run_* directory name. Record the mapping in a manifest so the audit trail stays explicit.

Step 4: Aggregate Each Recipe Separately

```
python aggregate_runs.py --runs-root runs\transfer_suite\rotating_opponents_holdout_endpoint
python aggregate_runs.py --runs-root runs\transfer_suite\rotating_plus_nemesis_novelty_replay
python aggregate_runs.py --runs-root runs\transfer_suite\rotating_plus_replay_aware_selection
```

Skip the third command if you did not run the optional arm.

Step 5: Run The Paired Causal Analysis

Required two-arm comparison:

```
python analyze_causal_transfer.py --baseline-recipe rotating_opponents_holdout_endpoint --recipe-root
runs\transfer_suite\rotating_opponents_holdout_endpoint --recipe-root
runs\transfer_suite\rotating_plus_nemesis_novelty_replay
```

Optional three-arm comparison:

```
python analyze_causal_transfer.py --baseline-recipe rotating_opponents_holdout_endpoint --recipe-root
runs\transfer_suite\rotating_opponents_holdout_endpoint --recipe-root
runs\transfer_suite\rotating_plus_nemesis_novelty_replay --recipe-root
runs\transfer_suite\rotating_plus_replay_aware_selection
```

The script writes:

- causal_transfer_summary.json
- causal_transfer_report.md
- causal_transfer_report.pdf

Step 6: Refresh The Novelty Review

After the new paired dataset is in place, rebuild the novelty packet on the relevant transfer roots:

```
python review_novelty_spikes.py --runs-root runs\transfer_suite\rotating_plus_nemesis_novelty_replay
--output-dir runs\transfer_suite\novelty_review_heavy_<timestamp>
python review_novelty_spikes.py --runs-root runs\transfer_suite\rotating_opponents_holdout_endpoint
--output-dir runs\transfer_suite\novelty_review_baseline_<timestamp>
python review_novelty_spikes.py --runs-root runs\transfer_suite\rotating_plus_replay_aware_selection
--output-dir runs\transfer_suite\novelty_review_replay_aware_<timestamp>
```

Interpretation Order

1. Per-environment paired transfer result
2. Overall paired transfer average
3. Per-archetype failure modes
4. Behavioral-shift versus code-novelty interpretation
5. Qualitative novelty review

Do not invert this order. The phase-4 question is causal interpretation of the transfer result, not a search for larger novelty numbers.